KERNEL

Knowledge Base

Company & AI Engineer Profile for the RAG in Action Workshop

Purpose

This document is the source knowledge base for the Kernel Knowledge Assistant demo. It contains structured information about SalahAldeen Yazan Salah AlNaimat, his AI engineering background, selected projects, Kernel, and the company's services and training programs.

Prepared for workshop demonstration
Kernel Knowledge Assistant
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PROFILE

1. About SalahAldeen

SalahAldeen Yazan Salah AlNaimat is an AI Engineer at Kernel with a multidisciplinary technical background spanning artificial intelligence, software development, computer networks, and cybersecurity.

He earned a Bachelor's Degree in Computer Engineering from The Hashemite University in Jordan. His academic background exposed him to programming, computer architecture, networking, security, and intelligent systems. Over time, his strongest professional interest became Artificial Intelligence and the engineering of practical AI systems.

SalahAldeen also completed practical training with multiple companies in the AI field. These experiences helped him move beyond academic exercises and work with real problems, business requirements, and implementation constraints.

Current areas of interest

- Machine Learning and Deep Learning
- Natural Language Processing and Transformer models
- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- AI Agents and Multi-Agent Systems
- AI automation and intelligent decision support
- AI safety, governance, and reliable system design

TECHNICAL BACKGROUND

2. Machine Learning Experience

SalahAldeen has practical experience with both classification and regression problems. One of his major Machine Learning projects was a Credit Card Customer Classification system developed using real banking data.

The project focused on analyzing customer information and identifying useful patterns for classification. Because the dataset came from a real banking context, the work required attention to data quality, preprocessing, feature engineering, exploratory analysis, model selection, and evaluation rather than simply training a model on a clean academic dataset.

> **Key learning**
> The project demonstrated how Machine Learning can support business decision-making when model development is combined with careful data preparation and evaluation.

TECHNICAL BACKGROUND

3. Deep Learning and NLP Experience

Deep Learning

SalahAldeen worked on a multi-class image classification project in which a Deep Learning model learned visual patterns and assigned images to one of several categories. The project involved image preprocessing, neural network training, validation, and performance evaluation.

Natural Language Processing

He also worked on a Fake News Classification project using BERT, a Transformer-based language model. The system analyzed textual news content and classified whether the content was likely to represent fake or legitimate news.

This project provided hands-on experience with NLP, text preprocessing, Transformer architectures, BERT, text classification, model fine-tuning, and semantic language understanding. It also created a strong foundation for later work with modern Large Language Models.

SELECTED PROJECT

4. AI Lead Qualification Multi-Agent System

One of SalahAldeen's practical AI projects was an AI Lead Qualification Multi-Agent System designed to automate the analysis of potential customers who contact a business through social media channels.

Instead of requiring employees to manually read every conversation, specialized AI agents analyze the customer's messages, infer intent and interest level, and classify the lead as Hot, Warm, or Cold.

When the system identifies a Hot Lead, it automatically sends an urgent bot notification to the responsible team so that high-priority customers can receive a fast response. The workflow also records incoming leads and relevant details in a structured Excel sheet without manual data entry.

> **Simplified workflow**
> Customer Message -> AI Agents -> Message Analysis -> Lead Classification -> Priority Decision -> Team Notification -> Automatic Lead Recording

The objective was to reduce repetitive manual work, improve response speed, and help the business team focus attention on leads with the highest conversion potential.

PROFESSIONAL ROLE

5. Current Role at Kernel

SalahAldeen currently works as an AI Engineer at Kernel. His role focuses on developing AI-powered systems and applying modern Artificial Intelligence technologies to real business and industry problems.

His current work includes Large Language Models, AI Agents, Multi-Agent Systems, Retrieval-Augmented Generation, prompt engineering, AI-powered decision support, intelligent automation, AI system architecture, AI safety and governance, and evidence-grounded AI systems.

A major part of this work involves designing systems that do more than generate text. These systems may retrieve external knowledge, analyze evidence, follow business rules and protocols, coordinate specialized AI components, and generate recommendations that support human decision-making.

CURRENT PROJECT

6. MABIEN - Healthcare Insurance AI

One of the major projects SalahAldeen currently works on at Kernel is MABIEN, an AI-powered solution for the Saudi healthcare sector. MABIEN is designed to support interactions between hospitals and health insurance companies, particularly during the processing and review of medical insurance claims and related decisions.

Healthcare insurance teams may need to review large volumes of medical requests and claims. A single case can require examination of medical information, coverage rules, clinical protocols, supporting evidence, and other relevant information before a decision can be made. When this process is repeated across thousands of claims, it becomes a significant operational challenge.

MABIEN uses a Multi-Agent AI architecture to automate and accelerate parts of this review process. Specialized agents analyze the available case information, evaluate it against supplied evidence and protocols, and generate decision-support outputs for insurance professionals.

> **Human-in-the-loop workflow**
> Medical Claim -> AI Agents -> Evidence and Protocol Analysis -> Decision Support -> Recommendation -> Human Review -> Approval or Rejection

The system is designed to support the insurance professional rather than replace the final human decision. The responsible employee reviews the AI-generated recommendation and remains responsible for the final approval or rejection according to the applicable process.

AI SAFETY

7. Reliability and Governance in MABIEN

Reliability is especially important when Artificial Intelligence is used in healthcare and insurance environments. For that reason, MABIEN is designed to avoid depending only on the general knowledge of an LLM.

The system is supplied with relevant evidence, protocols, and controlled knowledge sources. Governance mechanisms are used to reduce unreliable outputs, minimize hallucination risk, and improve decision consistency.

Important design principles

- Evidence-grounded decision support
- Protocol-based analysis

- Retrieval-Augmented Generation
- Multi-Agent coordination
- Controlled AI behavior and governance
- Hallucination reduction
- Decision consistency
- Traceable recommendations
- Human review and human-in-the-loop decision making

The objective is not only to make the review process faster, but also to provide useful, traceable, and reliable support for professionals handling high volumes of healthcare insurance cases.

COMPANY

8. About Kernel

Kernel is a technology company that provides Artificial Intelligence solutions, technical consulting, and technology-driven services for organizations in Saudi Arabia and the wider region. The company focuses on applying modern technologies to real operational and business challenges.

Kernel operates in both Jordan and Saudi Arabia. Its Jordan office is in Amman, near University Street, and its Saudi presence is based in Riyadh.

Kernel is part of Falak Investment in Saudi Arabia and operates as its technical arm, contributing to the design and delivery of technology and Artificial Intelligence solutions.

COMPANY

9. Clients and Industry Experience

Kernel has worked with major organizations and institutions on Artificial Intelligence and technology initiatives. Its experience includes engagements with the Saudi Ministry of Health, the Saudi Ministry of Defense, Monsha'at, and Airbus.

Across these engagements, Kernel has provided combinations of AI solutions, technical solutions, and consulting services. Working across government, healthcare, enterprise, and industrial environments gives the team practical exposure to complex requirements, operational constraints, and real implementation challenges.

This experience helps Kernel connect emerging Artificial Intelligence capabilities with practical enterprise use cases rather than treating AI as a standalone technology experiment.

COMPANY

10. Kernel Services and Solutions

Artificial Intelligence Solutions

Kernel develops AI-powered systems designed around specific organizational and business requirements. Solutions may involve Machine Learning, Large Language Models, Generative AI, Retrieval-Augmented Generation, AI Agents, Multi-Agent Systems, intelligent automation, and AI-powered decision support.

AI Consulting

Kernel supports organizations that want to understand where AI can create real value. This can include defining the business problem, evaluating AI approaches, designing the solution architecture, and supporting implementation.

Technical Solutions

Kernel also provides broader technical solutions based on customer and project requirements. The company emphasizes choosing technologies according to the problem being solved rather than applying AI when it does not provide meaningful value.

Intelligent Automation

Kernel uses AI to reduce repetitive and time-consuming work. Intelligent automation can analyze incoming information, classify cases, prioritize actions, notify teams, and record structured data with minimal manual intervention.

TRAINING

11. Kernel AI Training Programs

In addition to developing AI solutions and providing consulting services, Kernel delivers Artificial Intelligence training programs. The goal is to connect theory with real industry experience and help participants understand how AI is applied in professional environments.

Kernel currently provides Artificial Intelligence - Level 1 and Artificial Intelligence - Level 2. The Level 2 program has received strong interest, and due to demand, a new cohort is planned for September 2026.

The instructors bring project-based professional experience into the training environment. Depending on the course level, participants may be exposed to AI fundamentals, Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, AI Agents, and practical AI development workflows.

> **Training philosophy**
> Participants should learn not only how a technology works, but also how engineers apply it, evaluate it, and deal with real project constraints.

COMPANY & CAREER

12. Team, Approach, and Current Focus

Kernel Team

Kernel operates with a focused team of approximately 10 professionals and is led by Engineer Abdullah Al-Ghwairi, CEO of Kernel. The team combines technical expertise, practical project experience, and an understanding of real customer requirements.

Kernel's AI Approach

Kernel approaches Artificial Intelligence as a problem-solving tool rather than a trend. A typical AI initiative starts by understanding the business problem, the available information, the parts of the process that can benefit from AI, the appropriate architecture, the evaluation strategy, the required safeguards, and the points where human decision-making should remain involved.

SalahAldeen's Current Professional Focus

SalahAldeen's current direction is focused on AI Engineering and intelligent system architecture. His main interests include production-oriented RAG systems, AI Agents, Multi-Agent architectures, LLM applications, intelligent automation, AI orchestration, evaluation, reliable AI systems, governance, and enterprise Artificial Intelligence.

His experience has progressed from traditional Machine Learning and Deep Learning to Transformer-based NLP and, more recently, to complete AI systems that combine knowledge retrieval, LLMs, agents, tools, governance, and human decision-making.

End of Knowledge Base